

Jaehyun Choi

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PROFILE

Computer Science and Engineering undergraduate at Seoul National University, planning to pursue a minor in Mathematics. Experienced in C++ algorithmic problem solving and LLVM IR optimization, with competitive programming achievements at ICPC and UCPC. Interested in software engineering and quantitative trading.

EDUCATION

Seoul National University Mar. 2022 – Expected Feb. 2028
B.S. in Computer Science and Engineering Seoul, South Korea

- **GPA:** 4.26/4.3
- **Planned Minor:** Mathematics
- **Scholarship:** National Science and Engineering Scholarship

Seoul Science High School Mar. 2018 – Feb. 2021
High School Diploma Seoul, South Korea

PROJECTS

SWPP Project | LLVM IR Optimization Apr. 2026 – Jun. 2026

- Collaborated in a team of four to develop LLVM IR optimization passes in C++, including **FunctionInline**, **PostCleanup**, **BitwiseMaskToRem**, **GVNSimulate**, and **BitTestToDivRem**
- Refactored the function inlining pass to satisfy a strict **32-register constraint** while preserving program correctness
- Wrote LLVM IR (.ll) test cases for implemented passes and reviewed teammates' pull requests, discussing correctness and edge cases
- Used **Alive2** to reason about LLVM IR transformations and ranked **3rd out of 24 teams** in the project competition

HONORS & AWARDS

ICPC Seoul Regional | Bronze Medal, 9th / 84 teams 2025

UCPC | 4th Prize, 10th Place 2025

UCPC | 5th Prize, 12th Place 2026

AWS × Codetree Programming Contest (ACPC) | 7th Place 2026

SKILLS

Programming Languages: C++, Python, Java, Verilog, OCaml (basic)

Tools & Technologies: Git, LLVM IR, Alive2

Selected Coursework: Algorithms, Advanced Algorithms, Data Structures, Computer Architecture, Automata Theory, Modern Cryptography, Linear Algebra, Real Analysis